

Regime-Adaptive Portfolio Optimization via Forward-Filtered Hidden Markov Models

Drawdown Control Through Regime-Conditional Convex Optimization

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github.com/AbdelkrimCode/regime-adaptive-portfolio

July 21, 2026

Abstract

We present a regime-adaptive portfolio framework that combines a Gaussian Hidden Markov Model (HMM) for market regime detection with regime-conditional convex optimization. The key methodological contribution is the replacement of standard Viterbi decoding and forward-backward smoothing with a strictly causal forward filter — meaning the filter at time t uses only observations through t , eliminating within-window lookahead bias present in most HMM-based portfolio systems. Applied to eight liquid ETFs over 2006–2024, the strategy achieves a maximum drawdown of -27.0% versus -59.6% for SPY buy-and-hold, with a full-period Sharpe ratio of 0.617 versus 0.435. Eliminating lookahead bias comes at a small, honest cost: a controlled comparison holding the fitted models fixed shows the causal forward filter trailing forward-backward smoothed posteriors by 0.021 Sharpe (0.615 vs. 0.636) — the smoothed variant benefits from hindsight unavailable in live trading, so this gap is the expected direction, not a surprise, and the forward-filter number is the one that reflects what a live system could actually achieve. A frozen out-of-sample evaluation — training exclusively on pre-2019 data — yields a Sharpe of 0.518, confirming that performance does not depend on lookahead in the walk-forward procedure. In the held-out 2019–2024 period, the strategy’s Sharpe (0.453) trails SPY (0.677), consistent with a sustained equity bull market unfavorable to defensive allocation. Bootstrap significance testing yields $p = 0.266$; the strategy’s primary value proposition is tail risk control rather than return generation.

1 Introduction

Financial markets exhibit distinct statistical regimes — periods of low volatility trending growth, high volatility mean reversion, and acute crisis episodes — that are not well captured by time-invariant models. Portfolio strategies optimized under a single distributional assumption perform poorly when the underlying regime shifts, as evidenced by the catastrophic drawdowns of traditionally diversified portfolios during the 2008 Global Financial Crisis (GFC) and the 2022 simultaneous equity and bond selloff.

Hidden Markov Models [Hamilton, 1989] provide a principled framework for modeling latent market states. In the portfolio management literature, HMM-based regime detection has been

applied to asset allocation [Ang and Bekaert, 2002], volatility forecasting [Turner et al., 1989], and risk management [Guidolin and Timmermann, 2007]. However, most implementations rely on the Viterbi algorithm or forward-backward smoothing for state inference — both of which use future observations within the estimation window, introducing a subtle but consequential form of lookahead bias.

This paper uses “causal” in the signal-processing sense: the regime filter at time t conditions only on observations through t , not future observations. This is distinct from causal inference (do-calculus, interventional distributions); the contribution is methodological rather than causal-theoretic.

This paper makes the following contributions:

1. **Forward-only filter:** We replace Viterbi decoding and forward-backward smoothing with a strictly causal forward filter that computes $P(\text{state}_t \mid \mathbf{o}_{1:t})$ using only past observations, eliminating within-window lookahead bias. Holding the fitted HMM models fixed and varying only the decoding method, the causal forward filter trails smoothed posteriors by 0.021 Sharpe (0.615 vs. 0.636) and 2.8 percentage points of maximum drawdown. This is the expected direction of effect: smoothed posteriors have access to future observations within each estimation window and so can assign better-fitting labels in hindsight than any live system could achieve. The forward-filter number is the honest one — most published HMM-based portfolio backtests do not report this gap at all, and many implicitly rely on smoothing or Viterbi decoding without acknowledging the resulting lookahead.
2. **Posterior probability blending:** Rather than hard regime assignment, we blend optimizer outputs across all regimes weighted by filtered posterior probabilities, reducing turnover at transition boundaries.
3. **Nested cross-validation:** State count selection via BIC is performed independently at each quarterly fold, preventing information leakage from the full sample into model selection.
4. **Comprehensive validation:** We provide a frozen out-of-sample evaluation, feature ablation study, walk-forward leakage audit, bootstrap significance test, and stress tests — a level of methodological scrutiny absent from most published backtests.

The remainder of this paper is organized as follows. Section 2 describes the full methodology. Section 3 details the data and implementation. Section 4 presents performance results. Section 5 reports robustness analyses. Section 6 discusses limitations, and Section 7 concludes.

2 Methodology

2.1 Feature Construction

We construct three features from daily asset returns:

$$r_t^{\text{SPY}} = \log \left(\frac{P_t^{\text{SPY}}}{P_{t-1}^{\text{SPY}}} \right) \quad (1)$$

$$\sigma_t = \sqrt{\frac{252}{W_\sigma} \sum_{i=0}^{W_\sigma-1} (r_{t-i}^{\text{SPY}})^2} \quad (2)$$

$$\bar{\rho}_t = \frac{1}{N(N-1)} \sum_{i \neq j} \hat{\rho}_{ij,t} \quad (3)$$

where $W_\sigma = 21$ trading days, $\hat{\rho}_{ij,t}$ is the rolling $W_\rho = 63$ -day pairwise correlation between assets i and j , and $N = 8$ is the number of assets. Both window parameters are centralized in `config.yaml` and can be overridden via the sensitivity sweep. Features are standardized to zero mean and unit variance before HMM fitting.

2.2 Gaussian Hidden Markov Model

We model the feature sequence $\{\mathbf{x}_t\}$ as observations from a K -state Gaussian HMM with full covariance matrices:

$$P(\mathbf{x}_t \mid s_t = k) = \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \quad (4)$$

$$P(s_t = j \mid s_{t-1} = i) = A_{ij} \quad (5)$$

where $s_t \in \{1, \dots, K\}$ is the latent regime, $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}_k$ are regime-specific mean and full covariance, and A is the $K \times K$ transition matrix. Parameters $\{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, A, \boldsymbol{\pi}\}$ are estimated by the Baum-Welch algorithm (EM). Multiple random initializations ($n_{\text{init}} = 10$) are used to mitigate local optima; the initialization with the highest log-likelihood is retained, and each fold's fit is checked against its genuine EM convergence criterion (the log-likelihood improvement dropping below tolerance, as opposed to merely exhausting the iteration budget) rather than assumed. Across the current 75-fold walk-forward run, all 75 folds converged genuinely.

2.3 State Count Selection via BIC

The number of states K^* is selected at each quarterly fold by minimizing the Bayesian Information Criterion:

$$\text{BIC}(K) = -2\ell(\hat{\theta}_K) + p_K \log T \quad (6)$$

where $\ell(\hat{\theta}_K)$ is the maximized log-likelihood, T is the number of training observations, and p_K counts the free parameters assuming full covariance matrices:

$$p_K = \underbrace{K(K-1)}_{\text{transition}} + \underbrace{Kd}_{\text{means}} + \underbrace{K \frac{d(d+1)}{2}}_{\text{covariances}} + \underbrace{(K-1)}_{\text{initial state}} \quad (7)$$

with $d = 3$ features. For $K = 4$: $p_4 = 12 + 12 + 24 + 3 = 51$. Candidates $K \in \{2, 3, 4\}$ are evaluated independently at each fold, and the winning fit is reused directly rather than refit from scratch.

2.4 Causal Forward Filter

Standard HMM inference tools compute smoothed posteriors $P(s_t \mid \mathbf{x}_{1:T})$ using forward-backward passes over the full observation sequence. This introduces within-window lookahead: the regime label assigned to day t depends on observations $\mathbf{x}_{t+1}, \dots, \mathbf{x}_T$, which are unavailable in real-time.

We replace smoothing with a forward-only filter. The filtered posterior $\alpha_t(k) = P(s_t = k \mid \mathbf{x}_{1:t})$ is computed recursively:

$$\alpha_1(k) = \pi_k \cdot p(\mathbf{x}_1 \mid s_1 = k) \quad (8)$$

$$\alpha_t(k) = p(\mathbf{x}_t \mid s_t = k) \sum_{j=1}^K \alpha_{t-1}(j) A_{jk} \quad (9)$$

with normalization $\alpha_t \leftarrow \alpha_t / \sum_k \alpha_t(k)$ at each step to prevent underflow. The regime assignment on day t depends exclusively on $\mathbf{x}_{1:t}$. The initial state α_1 is computed using π learned on the training window; in test folds beginning mid-cycle, this initialization introduces a minor bias that cannot be corrected without knowing the true state at the fold start.

2.5 Walk-Forward Retraining

To prevent the HMM from using future market structure in regime detection, we employ quarterly walk-forward retraining with an expanding training window:

1. At each quarterly retraining date τ_q , fit the HMM on all data $\mathbf{x}_{1:\tau_q}$
2. Apply the fitted model via forward filter to observations in the subsequent quarter $[\tau_q + 1, \tau_{q+1}]$
3. Advance to the next quarter and repeat

Train and test windows are non-overlapping by construction. Data collection begins January 2005; the first 252 trading days are used to initialize rolling features and satisfy the minimum training window, so the backtest period begins January 2006. Of the 78 theoretical quarterly retraining dates spanning the data collection window, 3 are excluded because the expanding window does not yet meet the 252-day minimum training requirement, yielding 75 usable folds. A programmatic audit confirms zero leakage across all 75 folds (Section 5).

2.6 Label Permutation Handling

HMM states have no intrinsic semantic identity — state index 0 in quarter q need not correspond to the same market condition as state index 0 in quarter $q + 1$. We resolve this by assigning regime labels based on a risk-adjusted score: each state is ranked by $\text{mean_return} - 0.5 \times \text{mean_vol}$, with the lowest-ranked state labeled Crash, followed by Bear, Sideways, and Bull. The two inputs to this score are resolved by feature column name rather than position, so the ranking remains correct even when the feature set itself changes shape (Section 5 discusses a feature-ablation variant with no volatility column at all, where an earlier

position-based implementation silently substituted a different feature). This risk-adjusted ranking prevents mislabeling of high-volatility states with similar returns as safer regimes. We verify in Section 5 that regime rank ordering is stable across all quarterly folds, with no large drawdown coinciding with a label reassignment event.

2.7 Regime-Conditional Optimization

Four convex optimization programs are defined, one per regime. All use the Ledoit-Wolf shrinkage estimator [Ledoit and Wolf, 2004] for $\hat{\Sigma}$, applied to an expanding window of all returns through the retraining date.

Bull — Maximum Sharpe: Using the Lagrangian auxiliary variable reformulation [Cornuéjols and Tütüncü, 2006]:

$$\min_{\mathbf{y}} \quad \mathbf{y}^\top \hat{\Sigma} \mathbf{y} \quad (10)$$

$$\text{s.t.} \quad (\hat{\boldsymbol{\mu}} - r_f)^\top \mathbf{y} = 1, \quad \mathbf{0} \leq \mathbf{y} \leq 0.60 \cdot \mathbf{1}^\top \mathbf{y} \quad (11)$$

where $\mathbf{w} = \mathbf{y} / \mathbf{1}^\top \mathbf{y}$ and r_f is the time-varying 3-month T-bill rate ($\hat{\text{IRX}}$). The position-limit constraint is expressed in the transformed variable $\mathbf{y}$ so that, after the change of variables back to $\mathbf{w}$, it correctly enforces $w_i \leq 60\%$ regardless of the (a priori unknown) scale of $\mathbf{y}$.

Bear — Constrained Risk Parity: Weights approximately equalize each asset’s contribution to portfolio variance using the Spinu (2013) log-barrier surrogate [Spinu, 2013]:

$$\min_{\mathbf{w}} \quad \mathbf{w}^\top \hat{\Sigma} \mathbf{w} - \frac{1}{N} \sum_{i=1}^N \log w_i, \quad 0.01 \leq w_i \leq 0.60 \quad (12)$$

followed by post-hoc normalization $\mathbf{w} \leftarrow \mathbf{w} / \mathbf{1}^\top \mathbf{w}$. Critically, the budget constraint $\mathbf{1}^\top \mathbf{w} = 1$ is applied *after* solving, not as a constraint inside the optimization problem itself: the log-barrier objective’s first-order conditions already pin down each asset’s risk contribution up to a scale factor, and that property is scale-invariant under the post-hoc normalization. Including $\mathbf{1}^\top \mathbf{w} = 1$ as a hard constraint inside the same problem introduces a Lagrange multiplier that perturbs those first-order conditions and measurably loosens the equal-risk-contribution property this formulation is meant to deliver — on a controlled synthetic test, doing so widened the spread across assets’ risk contributions from approximately 0.002 percentage points (bounds-only, normalized after) to approximately 18 percentage points (equality-constrained). The lower bound $w_i \geq 0.01$ keeps the log-barrier term finite; the upper bound $w_i \leq 0.60$ matches the position limit applied to the other optimizers. The static Risk Parity benchmark uses the same formulation for comparability.

Sideways — Minimum Variance:

$$\min_{\mathbf{w}} \quad \mathbf{w}^\top \hat{\Sigma} \mathbf{w} \quad (13)$$

$$\text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad 0 \leq w_i \leq 0.60 \quad (14)$$

Position limits: All three optimizers above enforce $w_i \leq 60\%$ per asset. If any regime’s raw optimizer output nonetheless concentrates more than 99% of weight in a single asset — a numerical degeneracy rather than an intended allocation — it is replaced with an equal-weight fallback. For the Crash regime specifically, that fallback is scoped to the regime’s own

safe-haven subset $\mathcal{S}$ rather than the full 8-asset universe, since a full-universe fallback would inject equity exposure at precisely the moment the Crash regime exists to avoid it.

Crash — Inverse-Volatility Safe Haven: Weights inversely proportional to trailing volatility over the safe-haven subset $\mathcal{S} = \{\text{IEF}, \text{TLT}, \text{GLD}\}$:

$$w_i = \frac{\sigma_i^{-1}}{\sum_{j \in \mathcal{S}} \sigma_j^{-1}}, \quad i \in \mathcal{S} \quad (15)$$

where σ_i is the trailing volatility of asset i . Since IEF (7–10 year Treasuries) is typically the least volatile of the three safe havens, it typically receives the largest allocation of the three, with TLT and GLD receiving smaller allocations; the exact split varies with realized volatility at each retraining date. All optimizers use an expanding window of returns through the retraining date; early folds are dominated by 2008 volatility while late folds blend multiple regimes (see Section 6).

2.8 Posterior Probability Blending

Rather than committing to a single regime optimizer, daily weights are computed as a convex combination weighted by filtered posteriors:

$$\mathbf{w}_t = \sum_{k \in \mathcal{K}} \alpha_t(k) \cdot \mathbf{w}_t^{(k)} \quad (16)$$

where $\mathbf{w}_t^{(k)}$ is the weight vector produced by regime k ’s optimizer using all returns available through day t . Blending reduces turnover at transition boundaries and propagates the HMM’s regime uncertainty into the portfolio allocation. All four optimizer weight vectors are recomputed together whenever the dominant regime changes or at each quarterly retraining date, ensuring consistency across the blending.

3 Data & Implementation

3.1 Asset Universe

The investable universe consists of eight liquid US-listed ETFs spanning major asset classes:

Table 1: Asset Universe

Ticker	Description	Class
SPY	S&P 500	Equity
QQQ	NASDAQ-100	Equity
EFA	MSCI EAFE	Intl Equity
VNQ	US REITs	Real Estate
TLT	20+ Year Treasury	Long Bond
IEF	7-10 Year Treasury	Mid Bond
LQD	Investment Grade Corp	Credit
GLD	Gold	Commodity

Daily adjusted closing prices are sourced from Yahoo Finance via `yfinance`. Data collection begins January 2005 and runs through December 2024, yielding 5,032 daily price observations and 5,031 daily log returns per asset. Log returns are computed as $r_t = \log(P_t/P_{t-1})$.

3.2 Evaluation Design

Data collection begins January 2005; the first 252 trading days serve as the burn-in period for rolling feature initialization (21-day volatility, 63-day correlation) and to satisfy the minimum HMM training window. Accordingly, the backtest period begins January 2006 and runs through December 2024 — 18 years. The sample is split into a training period (January 2006 – December 2018) and a held-out test period (January 2019 – December 2024).

Walk-forward retraining uses a quarterly frequency with an expanding training window. Of 78 theoretical quarterly retraining dates, 3 are excluded due to the 252-day minimum window, yielding 75 usable folds. The 2019–2024 slice is used as the held-out evaluation period.

3.3 Transaction Costs

A flat round-trip cost of 2 basis points is applied to all portfolio weight changes, including the cost of the initial allocation from an unfunded (cash) starting position:

$$r_t^{\text{net}} = r_t^{\text{gross}} - c \cdot \sum_i |w_{t,i} - w_{t-1,i}| \quad (17)$$

where $c = 0.0002$ and $w_{0,i} \equiv 0$ for all assets at the start of the simulated series. Average daily turnover is 4.62%, implying an annualized transaction cost of approximately $4.62\% \times 252 \times 0.0002 \approx 0.23\%$ per annum. This is conservative for liquid ETFs at institutional scale and should be treated as a lower bound for retail implementation.

3.4 Benchmarks

We compare against five benchmarks, all net of the same 2bp transaction cost model:

- **SPY**: Buy-and-hold S&P 500
- **Equal Weight**: Monthly-rebalanced equal allocation across all 8 assets
- **60/40**: 60% SPY / 40% IEF, monthly rebalanced
- **Momentum**: Monthly rebalancing to the top-3 assets by equal-weighted trailing 252-trading-day return, computed through the most recent month end (no skip-month adjustment is applied)
- **Risk Parity**: Static constrained risk parity over all 8 assets, using the same log-barrier formulation as the Bear regime optimizer

3.5 Risk-Free Rate

The time-varying risk-free rate is sourced from the 3-month US Treasury Bill yield ($\text{\textasciitilde IRX}$ via Yahoo Finance). It is used both in the maximum-Sharpe QP objective and in all reported Sharpe ratio calculations, ensuring consistency between optimization inputs and performance reporting.

3.6 Implementation

The full pipeline is implemented in Python using `hmmlearn` [hmmlearn developers, 2023] (v0.3.3) for HMM estimation with full covariance matrices, `cvxpy` [Diamond and Boyd, 2016] for convex optimization, `scikit-learn` [Pedregosa et al., 2011] for Ledoit-Wolf shrinkage, and `joblib` for parallel fold execution. All code is available at <https://github.com/AbdelkrimCode/regime-adaptive-portfolio>. Continuous integration via GitHub Actions runs 111 unit and integration tests on every commit, including coverage for cache-consistency checks, EM convergence diagnostics, and the risk-parity and bootstrap corrections discussed in Sections 2 and 5.

4 Results

4.1 Full-Period Performance (2006–2024)

Table 2 reports annualized performance metrics over the full backtest period. The regime-adaptive strategy achieves a Sharpe ratio of 0.617 versus 0.435 for SPY buy-and-hold. The primary outperformance is in drawdown control: maximum drawdown of -27.03% versus -59.58% for SPY, and a Calmar ratio of 0.233 versus 0.140.

Table 2: Full-Period Performance (2006–2024)

Metric	Ours	SPY	EW	60/40	Momentum	RP
Ann. Return	6.28%	8.34%	5.96%	6.88%	5.63%	5.75%
Ann. Vol	8.09%	19.51%	11.17%	11.17%	13.28%	10.55%
Sharpe	0.617	0.435	0.448	0.525	0.388	0.454
Max DD	-27.03%	-59.58%	-35.05%	-35.60%	-27.39%	-32.84%
Calmar	0.233	0.140	0.170	0.193	0.206	0.175

The strategy’s volatility (8.09% annualized) is dramatically lower than all equity-heavy benchmarks, reflecting the regime-conditional allocation to bonds and gold during Bear and Crash periods. This lower volatility drives the Sharpe advantage despite lower raw returns.

4.2 Held-Out Test Period (2019–2024)

Table 3 reports performance on the held-out 2019–2024 period. The strategy’s Sharpe (0.453) lags SPY (0.677) significantly — the 2019–2024 period was characterized by a sustained equity bull market interrupted by brief, fast-recovering corrections, an environment unfavorable to regime-switching strategies that allocate defensively.

Table 3: Held-Out Performance (2019–2024)

Metric	Ours	SPY
Ann. Return	6.31%	14.87%
Ann. Vol	9.55%	19.91%
Sharpe	0.453	0.677
Max DD	-27.03%	-35.75%
Calmar	0.234	0.416

The maximum drawdown is identical in both the full period (Table 2) and the held-out period (Table 3). This is not a reporting error: the strategy’s peak-to-trough loss was realized during the 2020 COVID-19 crash, which falls within the 2019–2024 test window. The 2006–2018 training period contained no larger drawdown for this strategy.

4.3 Frozen Out-of-Sample Evaluation

To rule out lookahead in the walk-forward procedure itself, we train a single HMM on data through December 2018, freeze all parameters, and apply the model statically to 2019–2024 without any retraining. Table 4 compares frozen versus walk-forward performance.

Table 4: Frozen vs Walk-Forward (2019–2024)

Metric	Walk-Forward	Frozen	SPY
Ann. Return	6.31%	7.16%	14.87%
Ann. Vol	9.55%	9.54%	19.91%
Sharpe	0.453	0.518	0.677
Max DD	-27.03%	-27.79%	-35.75%
Calmar	0.234	0.258	0.416

The frozen model outperforms walk-forward (0.518 vs 0.453 Sharpe). This confirms the strategy generalizes — performance does not depend on retraining within the test period. At the same time, the frozen model’s outperformance may indicate that post-2018 regime structure differs from the 2006–2018 training distribution; quarterly retraining may be adapting to noise rather than signal in the test period.

4.4 Subperiod Analysis

Table 5 decomposes performance across three economically distinct subperiods. The periods 2006–2007 (pre-crisis) and 2010–2012 (recovery) are excluded from subperiod analysis; full-period performance is reported in Table 2.

Table 5: Subperiod Analysis

Period	Metric	Ours	SPY
GFC (2008–09)	Sharpe	0.387	-0.345
	Max DD	-10.90%	-56.38%
Bull (2013–19)	Sharpe	0.726	1.003
	Max DD	-7.54%	-19.82%
	Calmar	0.622	0.669
COVID+Rates (2020–24)	Sharpe	0.231	0.528
	Max DD	-27.03%	-35.75%
	Calmar	0.160	0.331

The GFC result is the most compelling: the portfolio gained +4.41% annualized versus SPY -15.49% , with maximum drawdown of -10.90% versus -56.38% for SPY, achieved without any forward-looking information. The Calmar ratio of 0.405 demonstrates positive risk-adjusted returns during the crisis. The strategy performs poorly in the COVID+rates period — the 2020 crash was too fast for quarterly retraining to adapt, and the 2022 environment (simultaneous equity and bond selloffs) violated the safe-haven assumption underlying the Crash optimizer.

4.5 Regime Distribution

Over the 2006–2024 backtest period (4,717 days with assigned regimes), the HMM assigns 1,264 days (26.8%) to Bull, 1,342 (28.5%) to Bear, 1,215 (25.8%) to Sideways, and 896 (19.0%) to Crash. Average regime duration ranges from 17.1 days (Bull) to 28.0 days (Crash), consistent with mean-reversion in market volatility. The empirical transition matrix shows Bull-to-Sideways as the single most common transition (60.8% of exits from Bull), with recoveries from both Bear and Crash tending toward Bull (49.3% and 21.9% of exits, respectively) or Sideways rather than a direct reversal.

We note that regime counts and downstream performance figures are not bit-reproducible across independent data downloads on different days: Yahoo Finance’s historical price history is occasionally revised retroactively (dividend and split adjustments can be backfilled), and because BIC-based state selection is a discrete choice, small price differences can occasionally shift which state count a fold selects. All figures in this paper come from a single, uninterrupted pipeline run for internal consistency; a different download session may reproduce the qualitative conclusions without matching every figure to the last decimal.

5 Robustness Analysis

5.1 Bootstrap Significance Test

We test whether the strategy’s Sharpe ratio exceeds SPY’s using a paired block bootstrap [Politis and Romano, 1994] with the same block offsets applied to both portfolio and SPY returns, preserving cross-series correlation. Under 1,000 bootstrap replications with block length $b = 20$ trading days, the empirical p -value is 0.266 — the null hypothesis that the

strategy’s Sharpe is no better than SPY’s cannot be rejected at conventional significance levels.

Sensitivity to block length: rerunning with $b = 10$, $b = 20$, and $b = 40$ produces p -values of 0.236, 0.266, and 0.276 respectively — stable across the range, confirming the result is not an artifact of block length choice.

This result is reported prominently because it is honest: the strategy does not statistically outperform SPY on a risk-adjusted basis. The value proposition is drawdown control and tail risk reduction, not Sharpe maximization. This p -value is computed by two independent implementations in the accompanying codebase (the main pipeline’s bootstrap module and a standalone stress-testing script); both currently agree to three decimal places, which we treat as a useful internal consistency check on the bootstrap procedure itself.

5.2 Walk-Forward Leakage Audit

We programmatically verify that no training fold observes future test data. For each of 75 quarterly folds, we record the last training date τ_{train} and the first test date τ_{test} . The condition $\tau_{\text{train}} < \tau_{\text{test}}$ holds for all 75 folds. Weekend and holiday offsets cause apparent one-to-two day gaps (e.g., training ends Friday, testing begins Monday), which are correct. Zero leakage is confirmed. The full audit is available at `results/walk_forward_audit.csv`.

We also verify that label permutation does not cause optimizer misrouting: regime rank ordering was stable across all 75 folds, with no quarter-to-quarter flip in the Crash/Bear/Sideways/Bull ordering. No large drawdown coincides with a label reassignment event.

5.3 Forward Filter vs Smoothed Posteriors

Table 6 quantifies the performance difference between the causal forward filter and forward-backward smoothed posteriors (loosely termed “Viterbi” in earlier drafts of this work, though the comparison uses `hmmlearn`’s smoothed posterior estimate rather than the literal Viterbi maximum-a-posteriori path). To isolate the effect of decoding method alone, both variants are computed from the *same* fitted HMM at each of the 75 folds — only the posterior computation differs, holding model fitting, feature construction, and optimization fixed.

Table 6: Forward Filter vs Smoothed Posteriors (Full Period 2006–2024)

Method	Full Sharpe	Max DD
Smoothed (forward-backward, uses hindsight)	0.636	-24.21%
Forward filter (causal)	0.615	-27.03%
Difference	-0.021	-2.82pp

The causal forward filter trails the smoothed variant by 0.021 Sharpe and 2.82 percentage points of maximum drawdown. This is the expected direction of effect, not a surprising one: smoothed posteriors at time t have access to observations $\mathbf{x}_{t+1}, \dots, \mathbf{x}_T$ within each fold’s test window, and can therefore assign a better-fitting regime label in hindsight than any real-time system could compute. The gap of 0.021 Sharpe is the quantified cost of removing that lookahead — smaller than we initially expected, which is itself informative: it suggests the forward filter’s uncertainty penalty is modest relative to the risk-adjusted state labeling and

position-limit constraints, which appear to do most of the work of protecting capital regardless of which decoding method supplies the regime probabilities. We report the forward-filter number throughout the rest of this paper, since it is the one a live system could actually achieve; a small, additional discrepancy between the forward-filter figure here (0.615) and the headline full-period Sharpe reported in Table 2 (0.617) arises because this controlled comparison uses a slightly simplified retraining-date convention, and is far smaller than the run-to-run variability documented in Section 4.

5.4 Feature Ablation Study

Table 7 reports full pipeline results across four feature configurations. The baseline (21-day vol, 63-day correlation) is not cherry-picked: `vol_10` produces comparable full-sample and held-out Sharpe to the baseline (0.670 full, 0.444 held-out), while `vol_42` achieves the highest full-sample Sharpe of any variant (0.767) but the lowest held-out Sharpe (0.103) — a clear overfitting signature rather than a genuinely better feature choice. `skew_kurt` remains the weakest held-out performer among the four, though a labeling implementation issue identified after the original version of this study was corrected: the regime-labeling heuristic (Section 2) previously resolved its two scoring inputs by column position rather than name, and this feature set has no volatility column, so it was silently scoring states on a skewness figure instead. With that corrected, `skew_kurt`’s held-out Sharpe is 0.260, materially better than previously reported though still the second-weakest variant. The baseline is retained for stability — it does not overfit in-sample and generalizes robustly, without `vol_42`’s fragility.

Table 7: Feature Ablation Results

Feature Set	Full Sharpe	OOS Sharpe	Max DD
Baseline (<code>vol₂₁</code> , <code>corr₆₃</code>)	0.617	0.453	-27.03%
<code>vol₁₀</code> (<code>vol₁₀</code> , <code>corr₆₃</code>)	0.670	0.444	-23.24%
<code>vol₄₂</code> (<code>vol₄₂</code> , <code>corr₆₃</code>)	0.767	0.103	-28.71%
<code>skew+kurt</code>	0.521	0.260	-26.11%

We additionally tested VIX and yield curve slope (5Y–3M Treasury spread) as supplementary macro features to reduce SPY circularity in an earlier iteration of this study; the qualitative finding was that these features improved full-sample fit but did not generalize out-of-sample, consistent with the HMM learning training-period-specific associations. We have not re-verified the exact magnitude of that result under the current, corrected pipeline and report it here only qualitatively; the 3-feature baseline remains the default.

5.5 Window Sensitivity Sweep

A 3×3 grid search over $\{10, 21, 42\}$ volatility windows and $\{42, 63, 126\}$ correlation windows yields full-sample Sharpe in the range $[0.50, 0.77]$ and held-out Sharpe in $[0.10, 0.53]$. The baseline configuration ($W_\sigma = 21$, $W_\rho = 63$) is not the in-sample optimum on either grid, but sits in a broadly stable region rather than an isolated peak, which is the more relevant robustness property than any single cell’s raw score.

5.6 Stress Tests

Table 8: Stress Test Results

Metric	Portfolio	SPY
VaR (95%, daily)	-0.79%	-1.84%
CVaR (95%, daily)	-1.19%	-3.07%
Sharpe, $VIX > 30$	-1.096	-1.952
MC Sharpe mean	0.619	—
MC Sharpe std	0.228	—
MC 95% CI	[0.187, 1.062]	—

Daily VaR and CVaR at the 95th percentile are less than half those of SPY, consistent with the drawdown control result. Both strategies produce negative Sharpe during $VIX > 30$ periods — crisis environments are adverse for all strategies — but the portfolio’s loss per unit of risk is substantially lower (-1.096 vs -1.952). Monte Carlo Sharpe (1,000 bootstrap paths, block-resampled to preserve autocorrelation) produces a mean of 0.619, closely matching the empirical estimate of 0.617, with no evidence of estimation bias.

5.7 PCA Regime Validation

Principal Component Analysis of the three features reveals that Bull, Bear, and Sideways regimes overlap heavily in the central region of PC1-PC2 space (39.3% and 33.3% variance explained respectively). Meaningful separation exists only in the left tail of PC1, corresponding to extreme negative returns and elevated volatility — Crash and severe Bear episodes.

This finding explains the non-significant bootstrap result: the HMM adds value primarily in tail events, not in distinguishing normal market conditions. Drawdown control is the correct framing for this strategy’s contribution. The heavy overlap in normal market conditions means the posterior blending in Eq. (16) is frequently combining near-equal weights across regimes rather than making confident allocations — this is a feature (smooth transitions) as much as a limitation.

6 Limitations

Gaussian emission assumption. Jarque-Bera tests reject normality for all four regimes ($p \approx 0$). Excess kurtosis is highest in the Crash regime (6.09) — clearly fat-tailed relative to a Gaussian — with Bull, Bear, and Sideways closer to 1.8–2.3. A Student- t HMM [Zhang et al., 2004] would be more appropriate, particularly for the Crash regime, but is not natively supported by `hmmlearn`; implementation from scratch via EM with t -distribution updates is left for future work.

Markov assumption. HMMs assume the current regime depends only on the previous regime. Markets exhibit momentum, mean-reversion, and long-memory volatility clustering (GARCH effects) at multiple horizons, violating the first-order Markov property. A higher-order HMM or regime-duration model would capture these dynamics at the cost of significantly increased parameter count.

SPY circularity. All three features are derived from SPY returns. The HMM detects regimes using the same instrument it indirectly trades, introducing circularity in the signal. Macro features (VIX, yield curve slope) were tested as orthogonal alternatives but did not generalize out-of-sample. True signal orthogonality would require data from a different domain (e.g., economic survey indices, options market microstructure).

Expanding covariance window. All optimizers use an expanding window of returns through the retraining date, growing from approximately 3,260 to 4,780 days over the backtest. Early folds are dominated by 2008 volatility; late folds mix this with the much calmer post-2022 regime. A rolling window would be more stationary but introduces estimation instability in short samples. The expanding window is retained as the default; rolling-window sensitivity is available in the repository.

Crash regime assumes safe-haven decorrelation. The inverse-volatility crash optimizer allocates to IEF, TLT, and GLD under the assumption that these assets are negatively or uncorrelated with equities during crises. In 2022, all three sold off simultaneously with equities due to the rate shock, limiting protection. A min-CVaR optimizer over the safe-haven subset would be more robust to correlation shocks.

Transaction cost model. The flat 2bp cost model does not account for market impact, bid-ask spread dynamics, or securities lending costs. For daily rebalancing at retail scale, realized costs would be higher. Sensitivity analysis over cost assumptions is available in the accompanying repository.

Time period bias. The 2006–2024 sample includes the GFC, a prolonged low-volatility bull market, COVID-19, and an aggressive rate cycle. The strategy has not been evaluated in structurally different environments such as the 1970s stagflation or prolonged deflation. Out-of-sample evidence from such periods is unavailable by construction.

Statistical significance. The paired block bootstrap p -value of 0.266 confirms that excess Sharpe over SPY is not statistically significant. Any claim of outperformance should be framed around drawdown control — a claim supported by the data — rather than risk-adjusted returns.

Sample mean returns as optimizer input. The Maximum Sharpe optimizer uses sample mean returns, which have near-zero signal-to-noise ratio at daily frequency. Shrinkage toward a factor model or Black-Litterman prior would be more robust; this is left for future work.

Risk parity remains an approximation. The Bear regime optimizer (Section 2) minimizes a log-barrier surrogate for equal risk contribution, subject to box constraints only, normalized post-hoc. A previous version of this optimizer additionally imposed the budget constraint inside the optimization problem itself, which we found materially loosens the equal-risk-contribution property (Section 2); with that constraint removed, the approximation is considerably tighter, but it remains an approximation rather than an exact solution.

Data is not bit-reproducible across download sessions. As discussed in Section 4, a fresh download of the same historical price series on a different day can shift regime counts and performance figures slightly, most likely due to Yahoo Finance backfilling minor historical corrections between sessions. Readers attempting to reproduce these results exactly should expect results representative of, rather than identical to, the figures reported here.

7 Conclusion

We presented a regime-adaptive portfolio framework built on a forward-filtered Hidden Markov Model and regime-conditional convex optimization. The central methodological contribution is the replacement of standard HMM smoothing with a strictly causal forward filter, eliminating the within-window lookahead bias present in most published HMM portfolio systems. Holding fitted models fixed, this correction costs 0.021 Sharpe relative to forward-backward smoothed posteriors (0.615 vs. 0.636) — the expected direction of effect, since smoothing has access to hindsight a live system would not — and the forward-filter figure is the one that reflects what such a system could actually achieve.

Applied to eight liquid ETFs over 2006–2024, the strategy achieves a maximum drawdown of -27.0% versus -59.6% for SPY buy-and-hold — a 54.6% reduction in peak-to-trough loss. During the 2008–2009 Global Financial Crisis specifically, the portfolio gained $+4.41\%$ annualized versus SPY -15.49% , with maximum drawdown -10.9% versus -56.4% for SPY. A frozen out-of-sample evaluation confirms these results are not an artifact of walk-forward lookahead: a model trained exclusively on pre-2019 data achieves Sharpe 0.518 on the 2019–2024 period, ahead of the walk-forward estimate of 0.453 on the same period.

The strategy does not statistically outperform SPY on a risk-adjusted basis (bootstrap $p = 0.266$), and underperforms significantly on raw returns in the 2019–2024 period — a sustained equity bull market unfavorable to defensive allocation. PCA analysis of the feature space confirms that regime separation is weak in normal market conditions and meaningful only in the left tail, explaining both the drawdown control result and the absence of Sharpe outperformance.

Future work includes replacing Gaussian emissions with Student- t distributions to better model fat-tailed returns, particularly for the Crash regime; incorporating Black-Litterman priors to improve expected return estimates; and extending the framework to higher-frequency data where the short-term predictive content of order book dynamics may provide more orthogonal signals for regime detection.

Acknowledgments

The author thanks the open-source maintainers of `hmmlearn`, `cvxpy`, and `yfinance` for the tools that made this implementation possible, as well as Mustapha Aziz Laroussi for inspiring this work.

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